

Assessing the credibility of the meta-analytic results

G0B75A Meta-analysis

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Outline

- a. Publication bias: assessment and correction
- b. Checking for outliers
- c. Checking assumptions

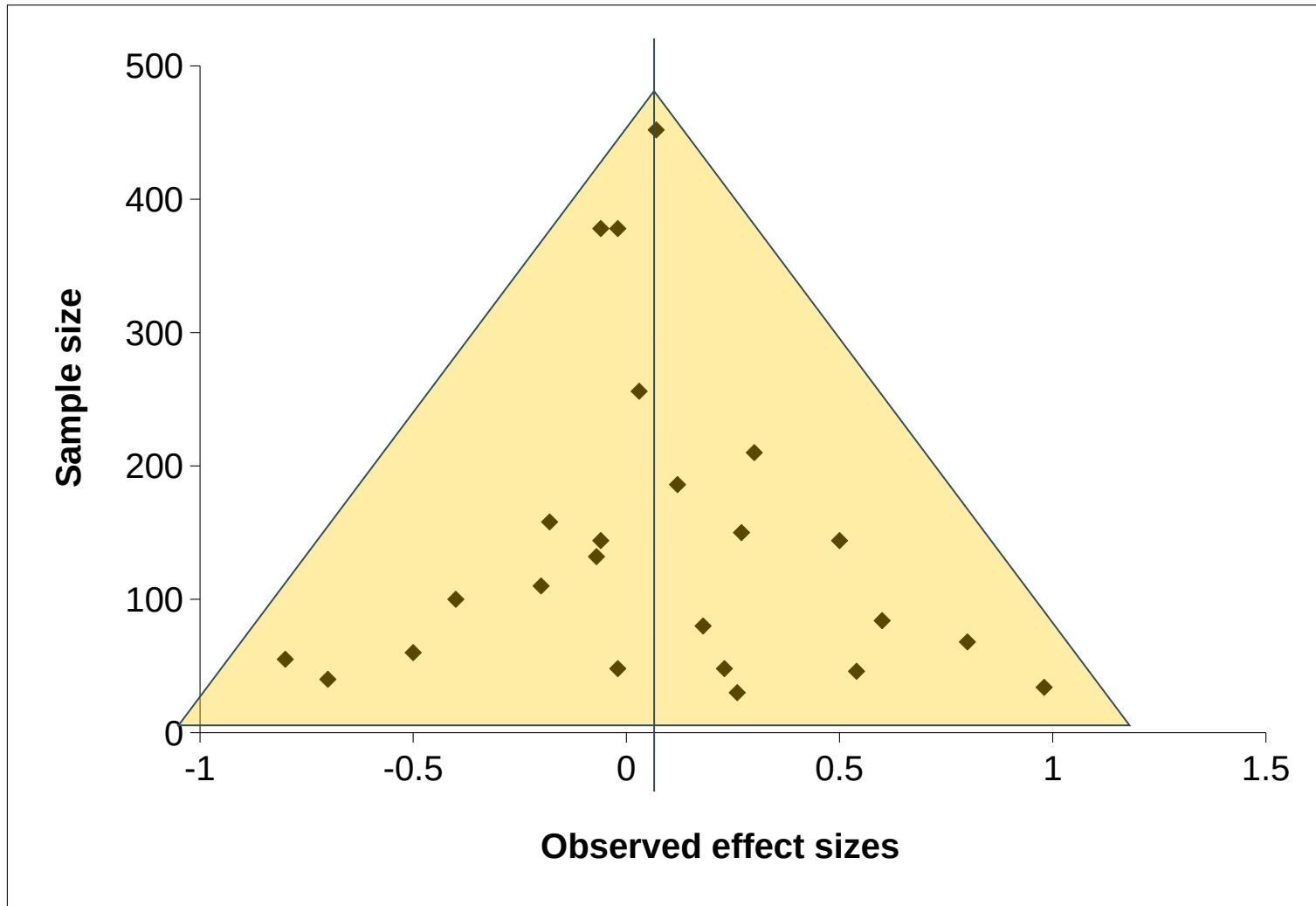
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Assessing publication bias

- Comparing overall effect size for published and unpublished results
- Funnel plot & Galbraith plot

The funnel plot (Light & Pillemer, 1984)



Example: Raudenbush & Bryk, 1985



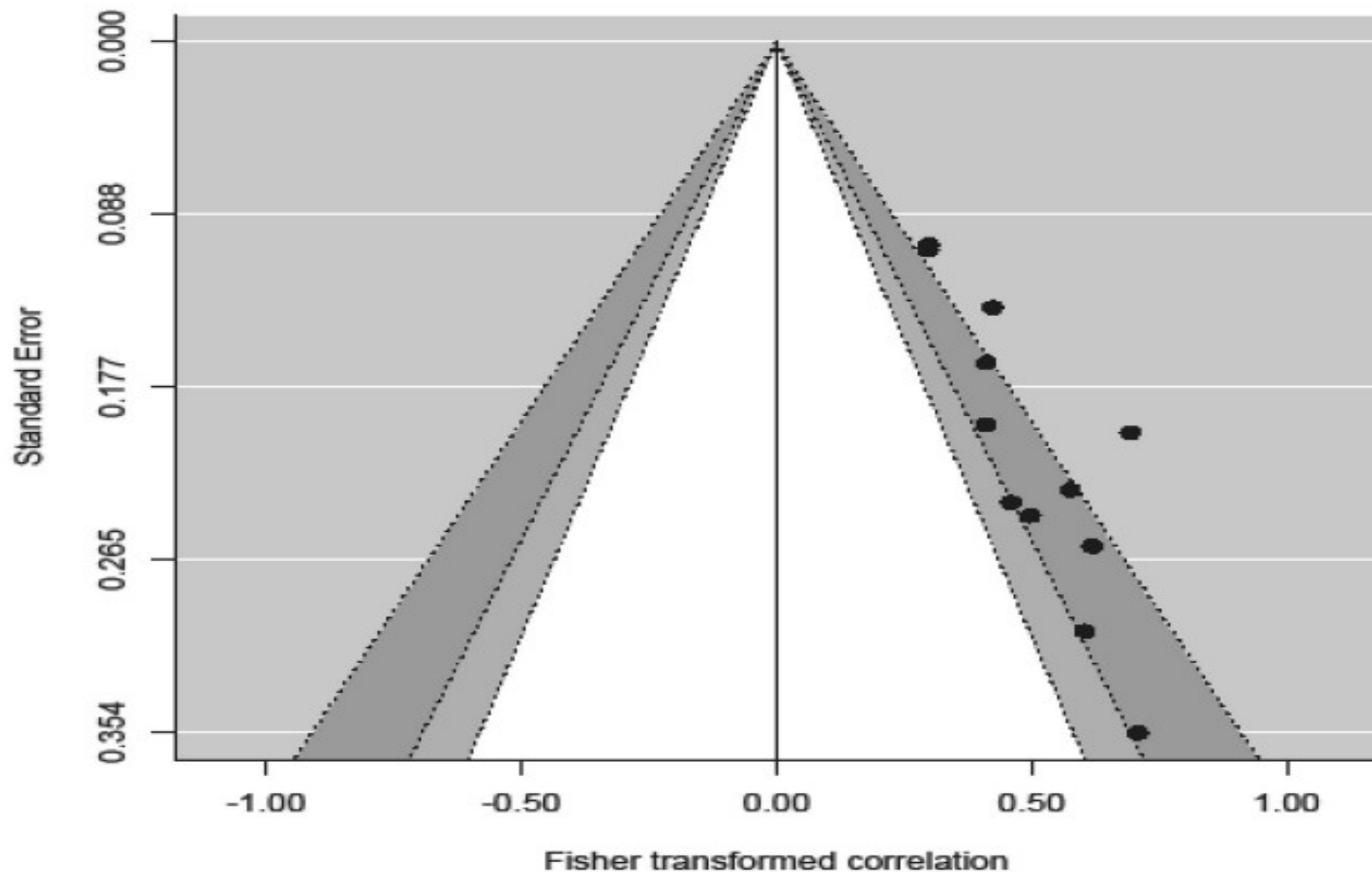
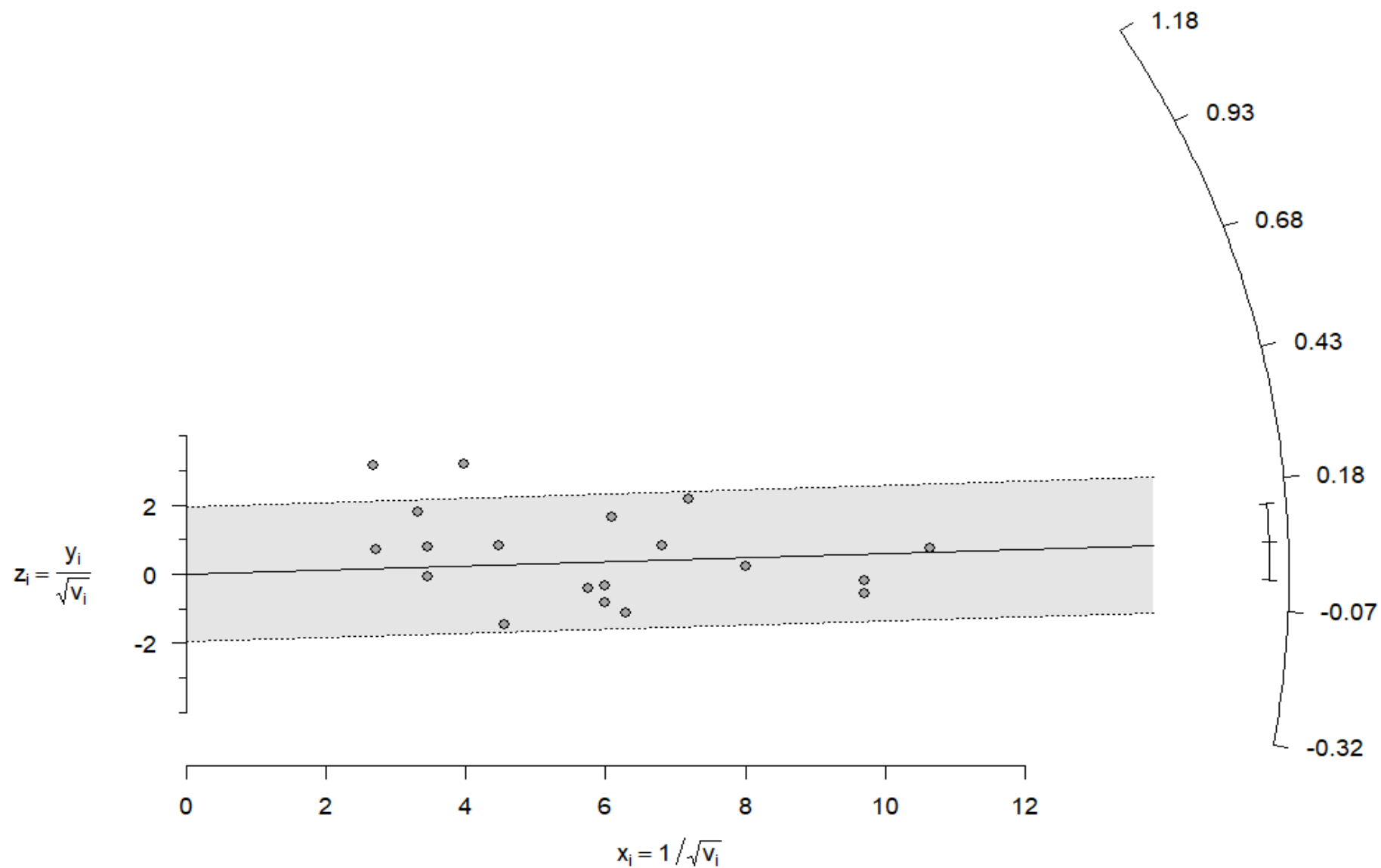


Figure 1. Contour-enhanced funnel plot of the meta-analysis of [McCall and Carriger \(1993\)](#). Areas represent studies with two-tailed p -values larger than .10 (white), smaller than .05 (light gray), smaller than .01 (dark gray), and smaller than .001 (light gray outside large triangle).

Radial plot (Galbraith, 1988)



Assessing publication bias

- Comparison of overall effect size for published and unpublished results
- Funnel plot & Galbraith plot
- Association between ES and N (or between ES and SE)
 - Kendall's tau
 - Spearman's rank correlation
- Include SE as a moderator variable (Egger's test)

But ... take care



- Negative correlation not necessarily due to publication bias!
- For some effect sizes, the SE and observed effect sizes are correlated.

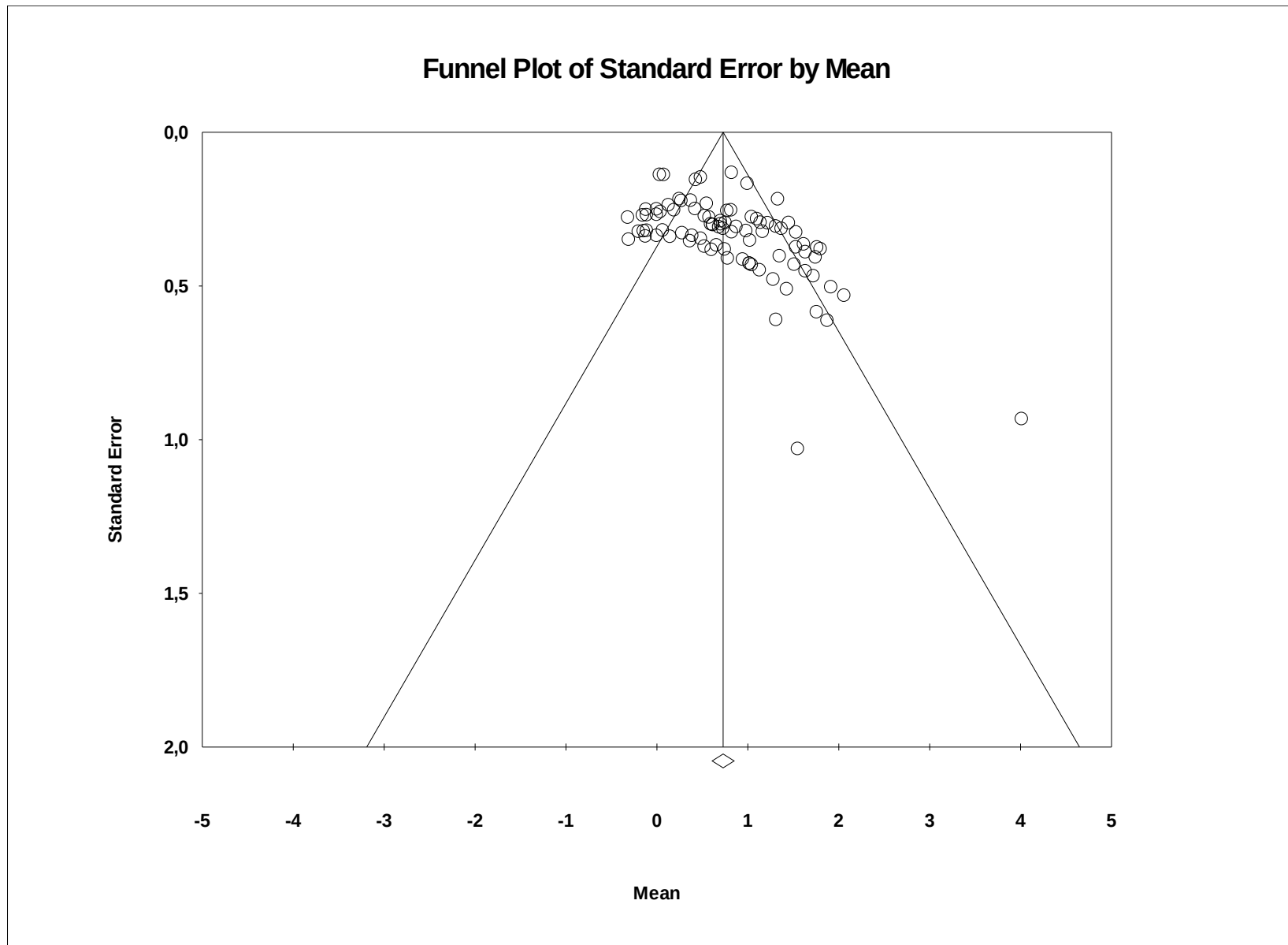
$$\text{E.g., } \hat{\sigma}_g^2 \cong \frac{n_E + n_C}{n_E n_C} + \frac{g^2}{2(n_E + n_C)}$$

- ⇒ Hedges & Olkin (1985) proposed a ‘variance stabilizing’ transformation of g . Improves results of Egger’s regression (Bertoldo, 2023)!
- ⇒ Similarly, Fisher’s z transformation makes the sampling variance independent of the effect size

Correcting for publication bias

- Trim & fill method

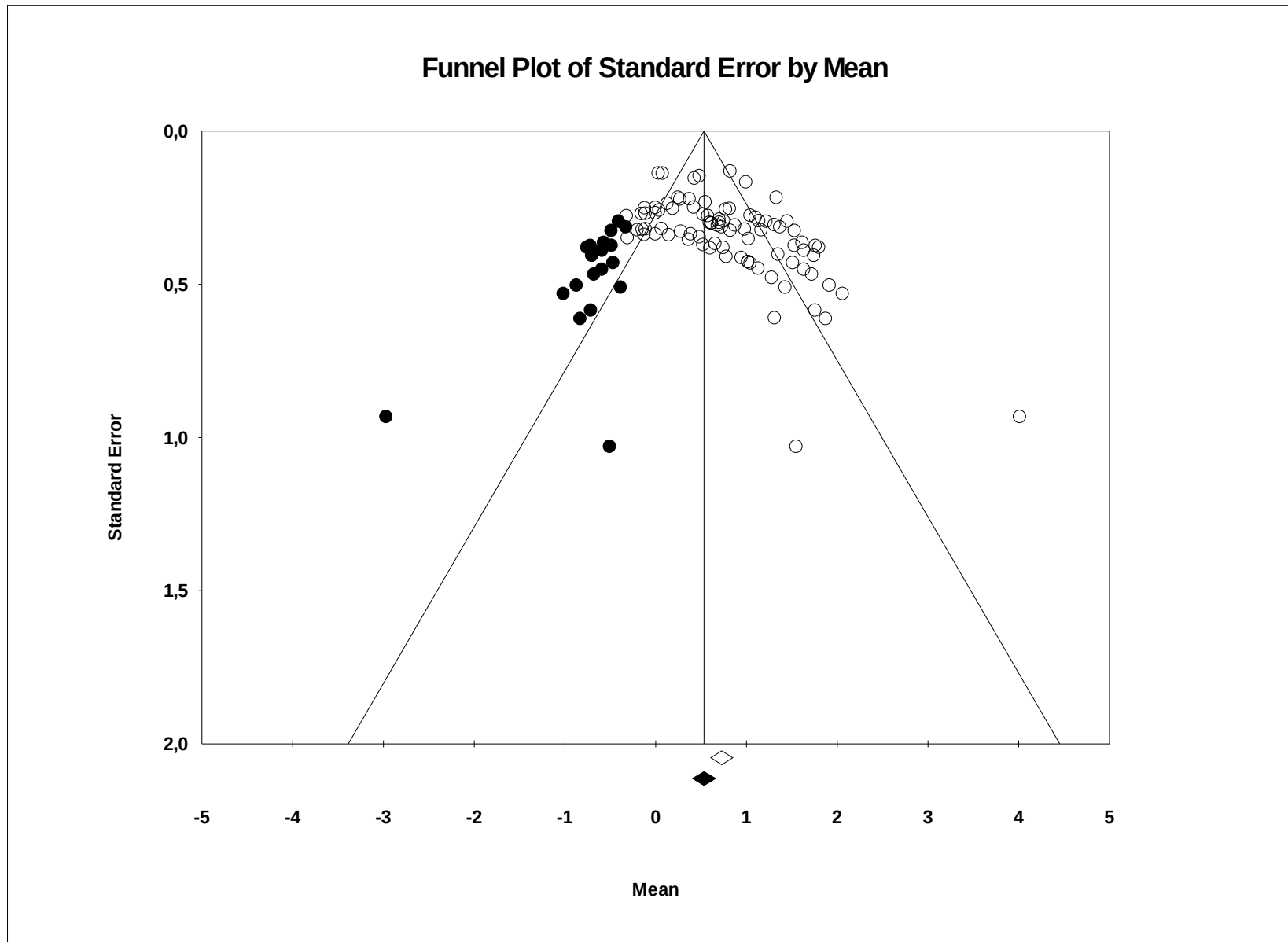
Trim & Fill method (Duval & Tweedie, 2000)



Trim & fill method

- Step 1: trims the asymmetric studies from the right-hand side to locate the unbiased effect (in an iterative procedure)
- Step 2: fills the plot by re-inserting the trimmed studies on the right as well as their imputed counterparts to the left of the estimated mean effect.

Trim & Fill method



Correcting for publication bias

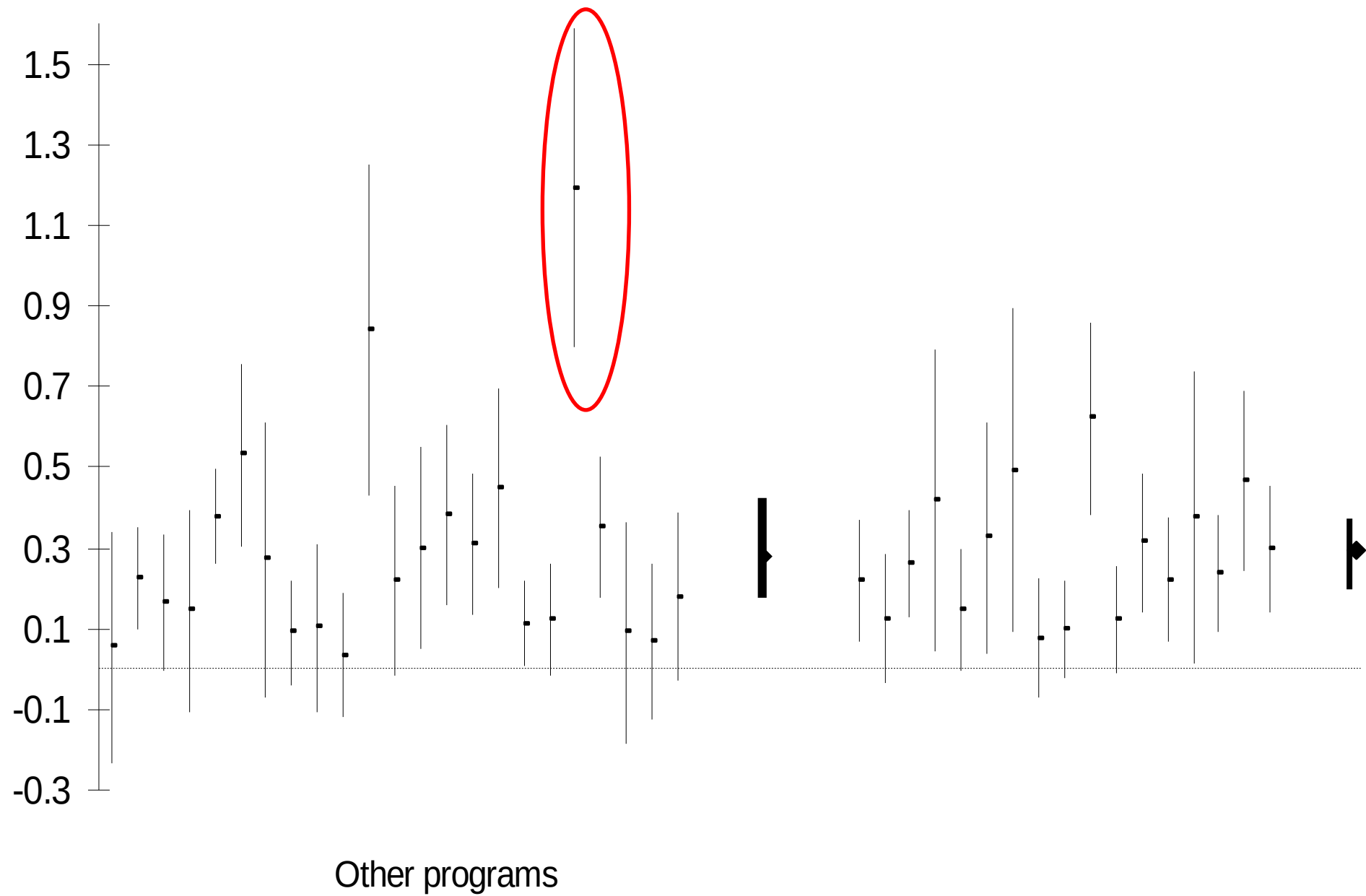
- Trim & fill method
- Include SE as a moderator variable (Egger's test)
- Selection models (e.g., Hedges & Vevea, 2005)
 - Idea: probability of publication depends on p -value
 - Model this probability (continuous or discrete step function; *a priori* or data-driven)
 - Weigh studies depending on the probability
 - But:
 - Complex
 - Risk of misspecification of the weight function
 - Requires many studies

Correcting for publication bias

- Trim & fill method
- Include SE as a moderator variable (Egger's test)
- Selection models (e.g., Hedges & Vevea, 2005)
- P-uniform (e.g., van Assen et al., 2015)
 - Idea: when testing $H_0: \theta = \theta$, $p \sim$ uniform distribution
 - Look at distribution of p -values when testing $H_0: \theta = \hat{\theta}$ (using studies with significant results only)
 - If distribution deviates from a uniform $\Rightarrow$ Indication of publication bias
 - Bias-corrected estimate: $\hat{\theta}$ for which distribution is close to uniform

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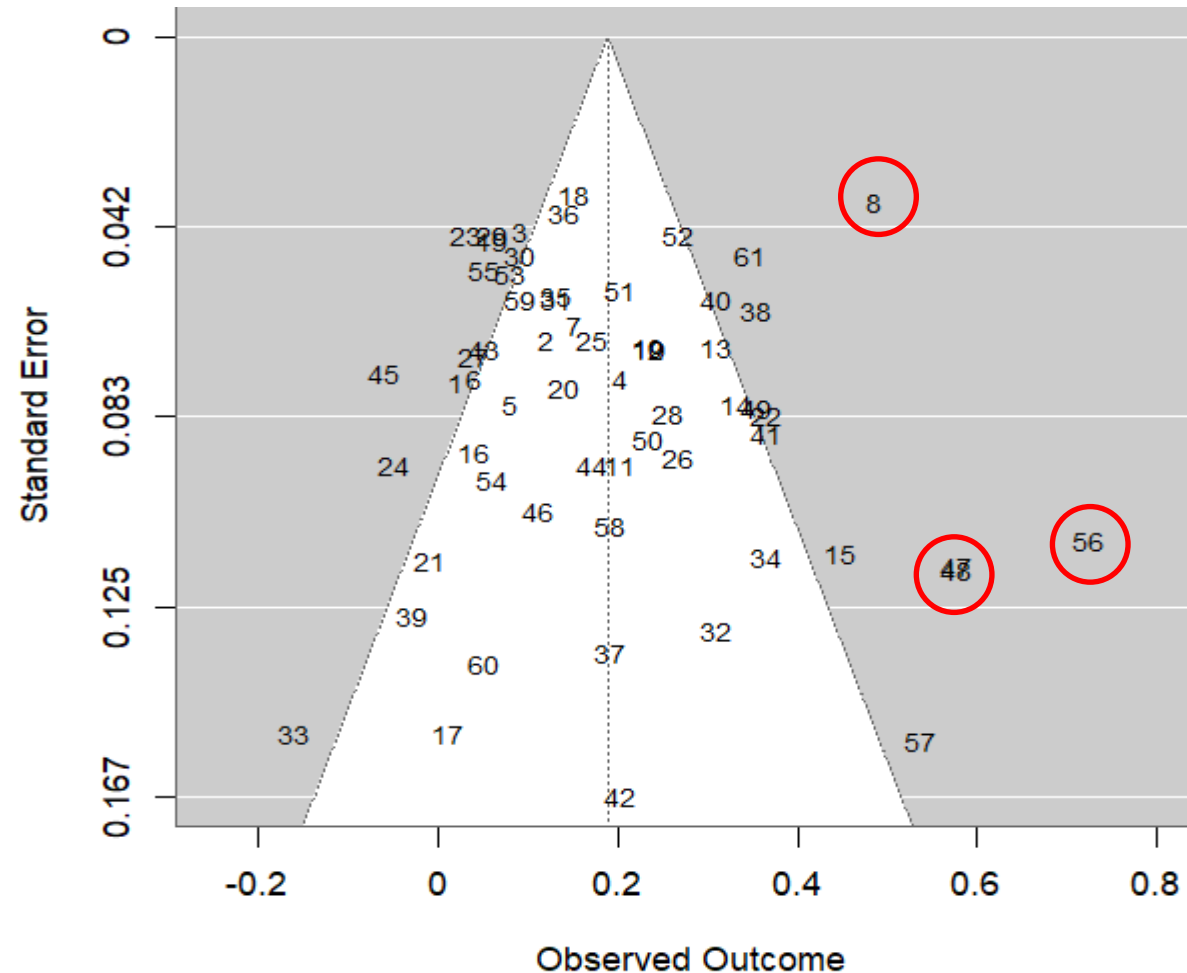


Geeraert, L., Van den Noortgate, W., Grietens, H., & Onghena, P. (2004). The effects of early prevention programs for families with young children at risk for physical child abuse and neglect. A meta-analysis. *Child Maltreatment, 9*, 277-291.

Detecting outliers

(Viechtbauer & Cheung, 2010)

- Visually (forest plot, funnel plot, Galbraith plot)

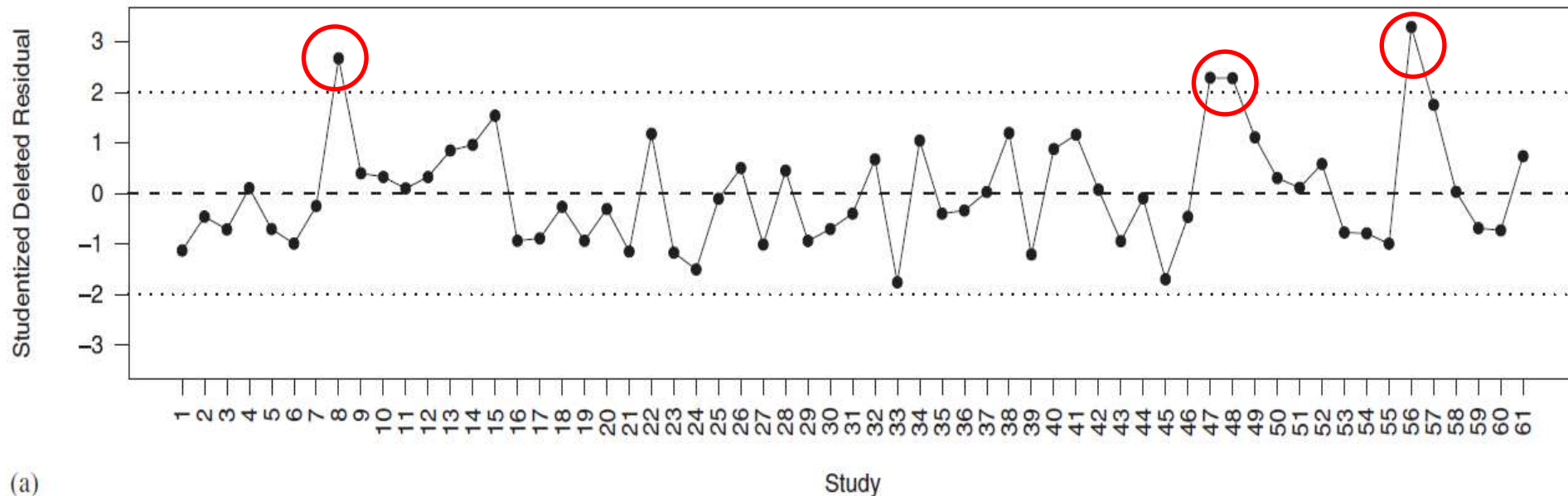


Detecting outliers

(Viechtbauer & Cheung, 2010)

- Using ‘studentized deleted residuals’

$$r_i = \frac{T_i - \hat{\theta}_{i(-i)}}{SE(T_i - \hat{\theta}_{i(-i)})}$$

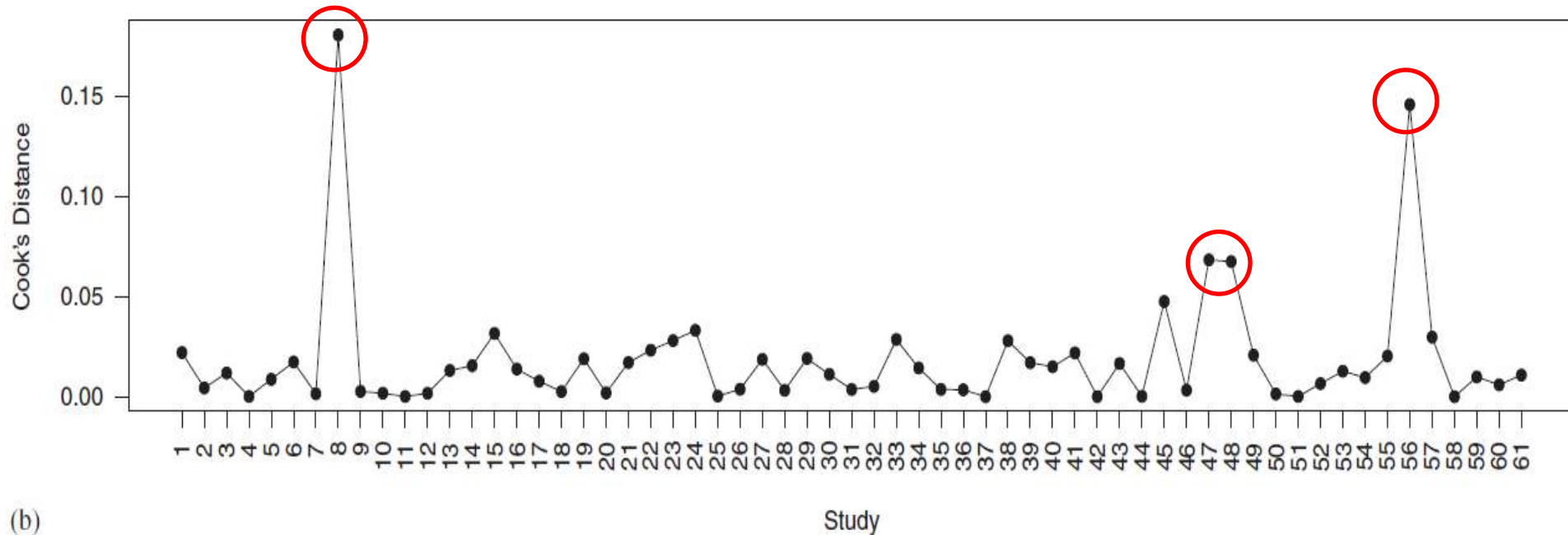


Detecting outliers

(Viechtbauer & Cheung, 2010)

- Using Cook's distance

$$D_i = \Sigma \frac{(\hat{\theta}_j - \hat{\theta}_{j(-i)})^2}{v_j + \hat{\tau}^2}$$

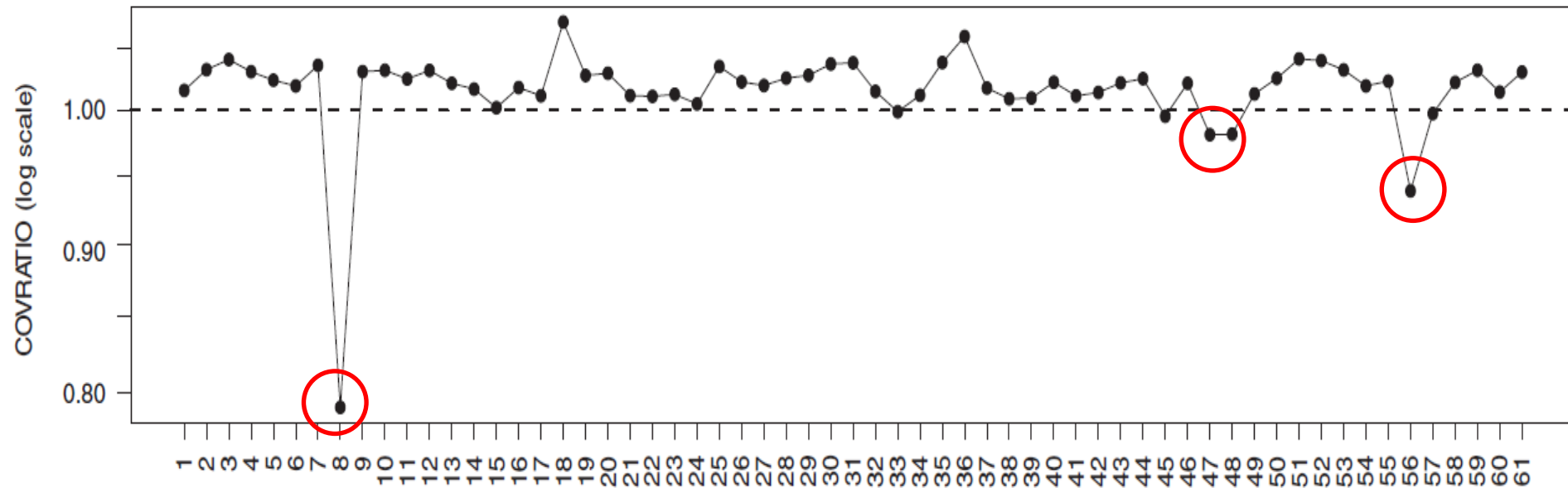


Detecting outliers

(Viechtbauer & Cheung, 2010)

- Using a covariance ratio

$$COVRATIO_i = \frac{Var(\hat{\theta}_{(-i)})}{Var(\hat{\theta})}$$



(c)

Study

Explaining outliers

- Errors in the calculation of effect sizes
- Errors in the data reported in the original studies
- Systematic differences between studies
- Coincidence

Sensitivity analyses

- Leave-one-out analyses
- Leaving out outlying values

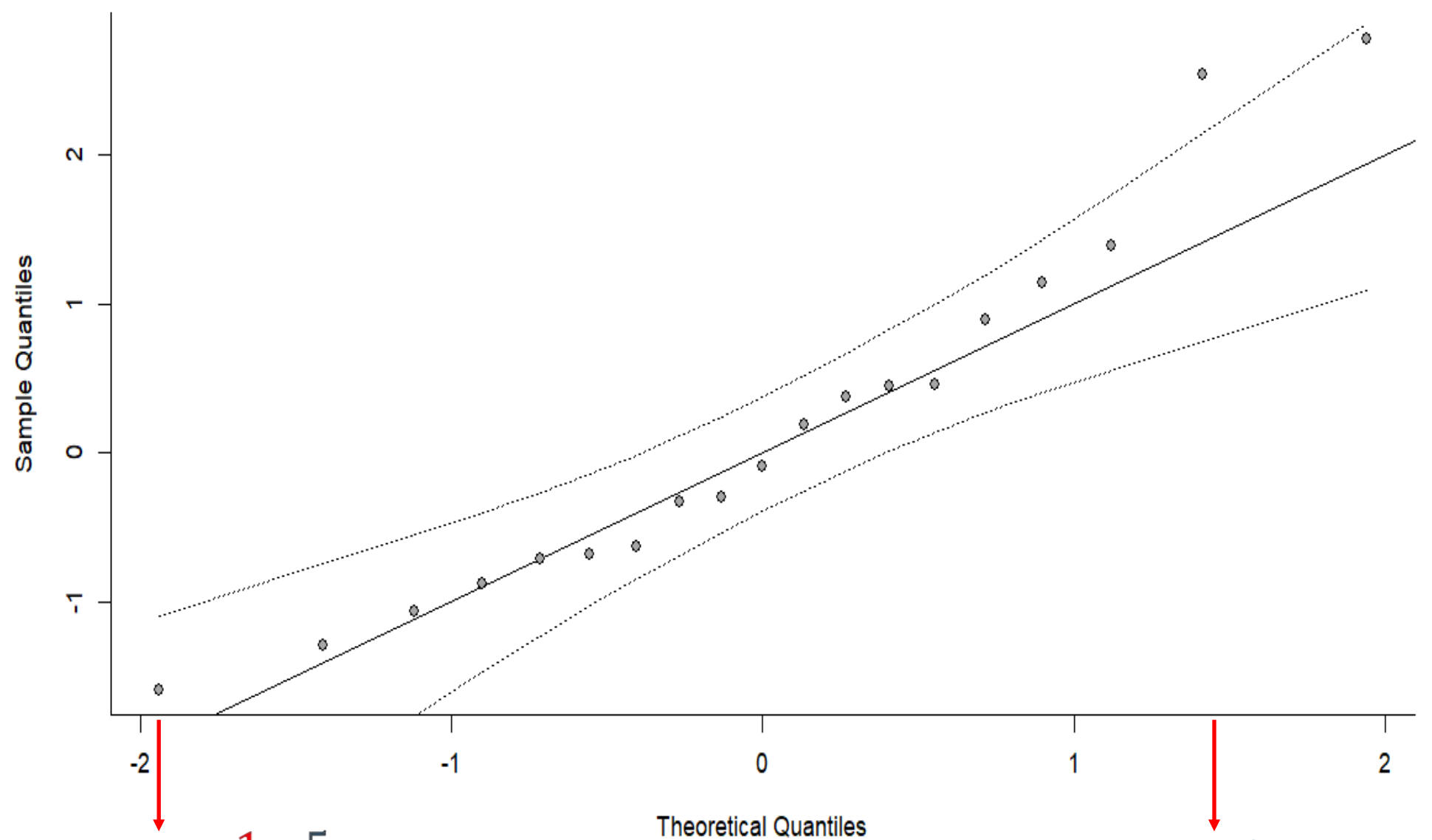
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Common assumptions:

- (Conditionally) common population effect size (FEM)
- Population effect sizes (conditionally) normally distributed (REM)

Normal Q-Q Plot



$$\Phi^{-1}\left(\frac{1-.5}{19}\right)=-1.94$$

$$\Phi^{-1}\left(\frac{18-.5}{19}\right)=1.41$$

Common assumptions:

- (Conditionally) common population effect size (FEM)
- Population effect sizes (conditionally) normally distributed (REM)
- Effect sizes are statistically independent (REM)
- Common between-study variance (REM)
- Number of studies relatively large (REM)
- Observed effect sizes unbiased (FEM/REM)
- Normal sampling distribution (FEM/REM)
- Retrieved studies as random sample (FEM/REM)
- Linear trends (meta-regression analyses)